



HOW TO

Make Your Own Stop-motion Animation

Fancy yourself the new Nick Park? Don't know who Nick Park is but want to make your own *Chicken Run*? You've come to the right place!

Nimish Chandiramani

We've done a lot of articles on making digital films, but we've always tactfully side-stepped this part: before you start reading, make sure you have some creative talent (or at least the desire to have some) handy. If you're unsure, ask someone who'll give you a straight answer. If their faces fall, it means you can safely turn the page, and we'll forgive you for it. Now, on to the matter at hand...

What It Is

Stop-motion animation is the art of making inanimate objects seem animated—by physically manipulating them, photographing them frame-by-frame and then stringing the frames together. Remember the AT-AT walkers in the *Star Wars* movies? Making them move was the work of stop-motion animators—very, very good ones.

What You Need

A plan: What's your scene going to depict? Do you need a specific setting?

The cast: Since you'll be animating the inanimate, anything that doesn't move of its own accord will work—your old G.I. Joe figures, a coin, scrunched up pieces of paper; if you want to animate your router's antennae twirling about, you can. You can even star in the video yourself for a quirky look.

The hardware: A respectable PC that can handle basic video editing should do fine—if your PC was bought less than a year and a half ago, it'll do. You'll also need a capture device like a webcam or camcorder.

The software: MonkeyJam—a little weird to use at first, but pretty neat after about ten minutes of use. Get it at www.giantscreamingrobotmonkeys.com/monkeyjam/ (love that URL).

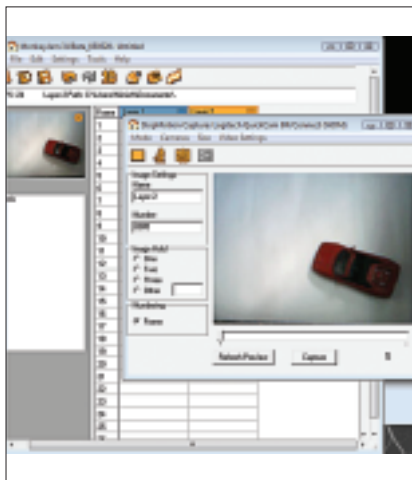
A set (optional): A big cardboard box.

Once you've placed all your characters and props, set up your webcam or camcorder for the shot and ensure that

it's not disturbed. Now, all that you need to do is start capturing frames and animating your characters.

How Monkeys Jam

In MonkeyJam, you'll be working with *Exposure Sheets* or, simply, Layers. When you start capturing frames, they're stored as frames in these layers—four frames per capture, to be precise. Hit [F6] to get to capture mode (Vista users: this causes Vista to unload Aero and use the Vista Basic theme—don't panic), and select Mode > Stop Motion. In this mode, you're presented with a slider at the bottom of your video so you can see what the previous frames looked like (to choose how many frames you want to



Use the slider to view preceding frames

see, make a choice under Image Hold) and compare it with the shot you're taking. This will help you ensure that your animation progresses smoothly.

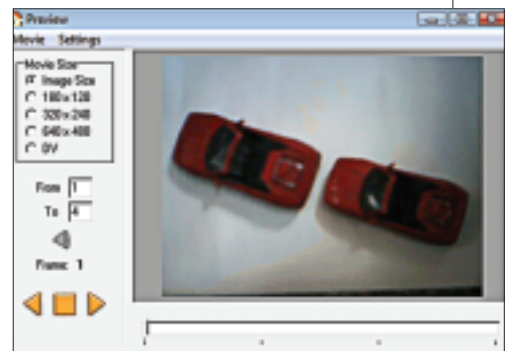
Preview

By default, MonkeyJam gives each capture three frames in the exposure sheet—you can change this by dragging the handle at the bottom of the captured image. Keeping images around for longer will lengthen the animation, but the transition to subsequent images will be painfully noticeable. On the other hand, keeping images restricted to two frames will result in smooth motion, but a very short video. You'll have to play with

these settings to strike a balance. To preview your movie so far, hit [F7].

The Shoot-the-last-frame-first Trick

To make sure your animation is progressing in the right direction, start by shooting the final pose of your character—this helps most with walking patterns,



Shoot the last frame first to make sure you're progressing properly—even for the silliest things

dances, and so on. Capture this image on to Layer 1, hit [Ctrl] + [L] to create a new layer and go to Composite > Blend. Now start capturing from the character's first position—all the captures should now go on to Layer 2. When you now preview this video, you'll be able to see the final frame blended over the animation—this will help you gauge whether you're approaching that final frame right and let you make corrections, if any. When you're done, just select Layer 1 and go to Edit > Delete Layer.

That's A Wrap!

One thing about stop-motion animation—it's a lot harder than it seems. It takes 24 frames to create a second of video—that's 1,440 frames for just one minute! Stop-motion animation software is usually nice and simple like MonkeyJam or Animator DV (find the free edition on the August 2007 DVD)—you'll need to follow up with a more fleshed-out video editor to work on it more. If you get around to making an animation of your own, send it in and we'll feature it on our DVD—get cracking! □

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